

YUVARAJ S

B.Tech Artificial Intelligence and Data Science

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CAREER OBJECTIVE

Motivated B.Tech Artificial Intelligence and Data Science student with hands-on experience building AI-driven applications using LLM APIs, RAG pipelines, prompt engineering, multimodal retrieval, and agentic AI. Proficient in Python with practical experience in FastAPI, vector similarity search, and multi-agent workflows. Seeking an AI/GenAI Engineer opportunity to build innovative, reliable, and production-ready AI solutions.

EDUCATION

HigherSecondary Education

2022-2023 | Score:69%

Government Higher Secondary School, Vathalmalai, Dharmapuri

Bachelor of Technology

2023-2027 | CGPA: 6.52

Bannari Amman Institute of Technology, Sathyamangalam, Erode

TECHNICAL SKILLS:

Programming Languages: Python, Java, C

AI / LLM: LLM APIs, Prompt Engineering, RAG Pipelines, Multimodal Retrieval, NLP, Agentic AI & Multi-Agent Systems

Frameworks & Tools: FastAPI, CrewAI, Git & GitHub

Databases: MySQL, MongoDB

AI / Retrieval: Vector Similarity Search, Multimodal Retrieval

Web Technologies: HTML5, CSS3, JavaScript

Core CS: Data Structures and Algorithms, OOPS, DBMS

PROJECTS:

Voice-Based AI Study Assistant

- Developed a voice-enabled AI study assistant capable of analyzing spoken input and extracting study topics automatically.
- Implemented interactive quiz and assessment features to improve learning engagement.
- Enabled real-time audio communication using speech processing techniques.

Visual Search Engine for Jewelry using Multimodal RAG

- Built a Retrieval-Augmented Generation pipeline integrating LLM APIs to generate product descriptions from retrieved jewelry data
- Designed vector similarity search across image, sketch, and text inputs for accurate multimodal retrieval
- Structured prompts and parsed LLM outputs to produce consistent, structured product descriptions

PlacementPal - AI Powered Placement Preparation Platform

- Built a multi-agent AI platform using LangGraph, CrewAI, and LangChain to generate personalized study plans and company-specific interview insights.
- Implemented a RAG pipeline with ChromaDB for semantic document retrieval, backed by FastAPI and asynchronous MongoDB persistence.
- Integrated the Tavily Search API for real-time company and hiring insights within a two-phase, stateful LangGraph workflow.

CropCopilot – AI-Powered Agricultural Intelligence System

- Integrated NVIDIA NIM LLM APIs with CrewAI agents to combine RAG-based knowledge retrieval with Text-to-SQL querying.
- Built and deployed a FastAPI backend serving real-time AI-generated insights with robust API and model error handling.
- Developed a proof of concept combining structured database queries with unstructured knowledge retrieval for agricultural decision support.

CERTIFICATIONS

Prompt Engineering — Navigate Labs

- Completed a 4-week intensive hands-on training covering prompt design techniques, few-shot and chain-of-thought prompting, and structuring inputs for reliable LLM outputs.

Retrieval-Augmented Generation (RAG) — Navigate Labs

- Completed a 4-week intensive hands-on training covering RAG pipeline design, vector embeddings, and semantic retrieval for grounding LLM responses in external knowledge sources.

Voice Artificial Intelligence — Navigate Labs

- Completed a 4-week intensive hands-on training covering speech-to-text and text-to-speech pipelines, real-time audio processing, and building voice-driven conversational AI systems